← Back **Life on the Early Earth**
Graded Quiz • 30 min

**Due** Mar 16, 11:59 PM CET

## Congratulations! You passed!

**Grade received** 95%

**Latest Submission Grade** 95%

**To pass** 80% or higher

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**1.** When the gas cloud from which a solar system forms collapses into a disc, what do we call the bright object in the centre? — 1 / 1 point

- star
- planet
- disc
- (●) protostar

Correct
A protostar is what we call the young star before it has begun to produce energy through nuclear fusion

**2.** What is the energy generating process in stars? — 1 / 1 point

- combustion
- (●) nuclear fusion
- dissolution
- nuclear fission

Correct
Nuclear fusion occurs when two or more atomic nuclei combine to form a new type of nuclei. Energy from this type of reaction is what fuels the sun.

**3.** What do we call the disc of material which orbits around a new star from which planets form? — 1 / 1 point

- Inner planets
- Solar system
- (●) Protoplanetary disc
- Outer planets

Correct

**4.** Do rocky planets form closer or further away from the star than gaseous planets? — 1 / 1 point

- Further away
- (●) Closer

Correct
It is suggested that the solar wind was stronger when the sun was younger. When this wind started up it would blow away light gaseous elements like hydrogen and helium away from the inner solar system. The wind would be less strong at the orbit of the outer planets and the gases would not be expelled past this. The gravity of the planets would allow them to mop up these elements.

**5.** What age are the rocks of the Isua region, some of the oldest rocks preserved on Earth? — 1 / 1 point

- 540 million years old
- 2 billion years old
- 4.6 billion years old
- (●) 3.5 - 4 billion years old

Correct
This is correct.

**6.** What evidence is seen in the Isua rocks which indicate water was present when they were laid down? — 1 / 1 point

- [x] sandstones

Correct
Sandstones can be lain down by the erosion and subsequent deposition of rock by water bodies such as rivers.

- [x] pillow lavas

Correct
Pillow lavas form when volcanic eruption occur under water.

- [x] limestones

Correct
Today limestones form in shallow marine settings

- [ ] shells

**7.** Which of these gases are thought to have assisted in keeping the Earth's temperature above freezing when the sun was not burning as brightly as today? — 1 / 1 point

- [x] methane

Correct
Methane is a greenhouse gas which means its presence in the atmosphere helps to trap heat.

- [ ] oxygen
- [x] carbon dioxide

Correct
Carbon dioxide is a greenhouse gas which means its presence in the atmosphere helps to trap heat.

- [ ] nitrogen

**8.** Which of these were hazards for life on the early Earth? — 1 / 1 point

- [ ] freezing temperatures
- [ ] forest fires
- [x] asteroid and comet impacts

Correct
Asteroid and comet impacts were more common early in the history of the Earth

- [x] high UV radiation

Correct
Because there was no ozone layer on the early Earth more UV radiation could pass through the atmosphere

**9.** What evidence do we have that the 'Late Heavy Bombardment' occurred? — 1 / 1 point

- (●) Many similarly aged impact craters on the moon
- Many similarly aged impact craters on the Earth
- A gap in the rock record.
- A mass extinction event

Correct
Because the moon is not tectonically active it has recorded the history of impacts in the vicinity of the Earth. A large number of impacts at a given time on the moon is likely to correlate with increased impacts on the Earth.

**10.** What percentage of Earth's history is not preserved in the rocks? — 1 / 1 point

- (●) 11%
- 1%
- 4%
- 25%

Correct
The rock record stretches back to about 4 billion years ago. The earth formed around 4.56 billion years ago.

**11.** What is the oldest fossil evidence we have for life on Earth? — 1 / 1 point

- 3.8 billion years old
- (●) 3.5 billion years old
- 540 million years old
- 2 billion years old

Correct
The oldest reported microfossils are from stromatolites dated at around 3.5 billion years old.

**12.** In what environment do stromatolites form today? — 1 / 1 point

- Rivers
- (●) Shallow marine
- Hydrothermal vents
- Deep ocean

Correct
Stromatolites are laminated mounds of sediment and syanobacterial mats. they form today in shallow marine settings.

**13.** Which stable form of carbon will life preferentially use? — 1 / 1 point

- Carbon 11
- Carbon 14
- Carbon 13
- (●) Carbon 12

Correct
Carbon 12 is one of two stable forms of carbon found in nature. It is preferentially used by life because it is lighter than carbon 13.

**14.** Which of these are problems with the evidence for the earliest life? — 0 / 1 point

- [ ] They have never been published
- [ ] Rocks are usually heavily metamorphosed
- [x] Observed features also explained by abiogenic processes

Correct

- [ ] Geological settings uncertain or disputed

You didn't select all the correct answers

**15.** Which organisms do the microfossils from the stromatolites in Western Australia most closely resemble? — 1 / 1 point

- worms
- (●) cyanobacteria
- fish
- soft-bodied animals

Correct

**16.** What is phylogenetics? — 1 / 1 point

- A period of geological time
- (●) The study of the evolutionary relationships between groups of organisms
- A species of organism
- The study of life in extreme environments

Correct

**17.** What is the name given to the organism which is proposed to be at the base of the tree of life from which all of the species we see today evolved? — 1 / 1 point

- Eukaryote
- Bacteria
- (●) Last Common Ancestor
- Slime mould

Correct

**18.** Which of these organisms are prokaryotes? — 1 / 1 point

- [x] green filamentous bacteria

Correct
Bacteria are prokaryotes

- [ ] microsporidia
- [x] cyanobacteria

Correct
Bacteria are prokaryotes (they have no cell nucleus)

- [ ] fungi

**19.** Which of these are two proposed alternatives to a tree of life? — 1 / 1 point

- [ ] String of life
- [x] A ring of life

Correct
This is correct

- [x] A web of life

Correct
This is correct

- [ ] Circle of life

**20.** Which type of organism is not a Eukaryote? — 1 / 1 point

- (●) Proteobacteria
- plants
- mammals
- slime molds

Correct